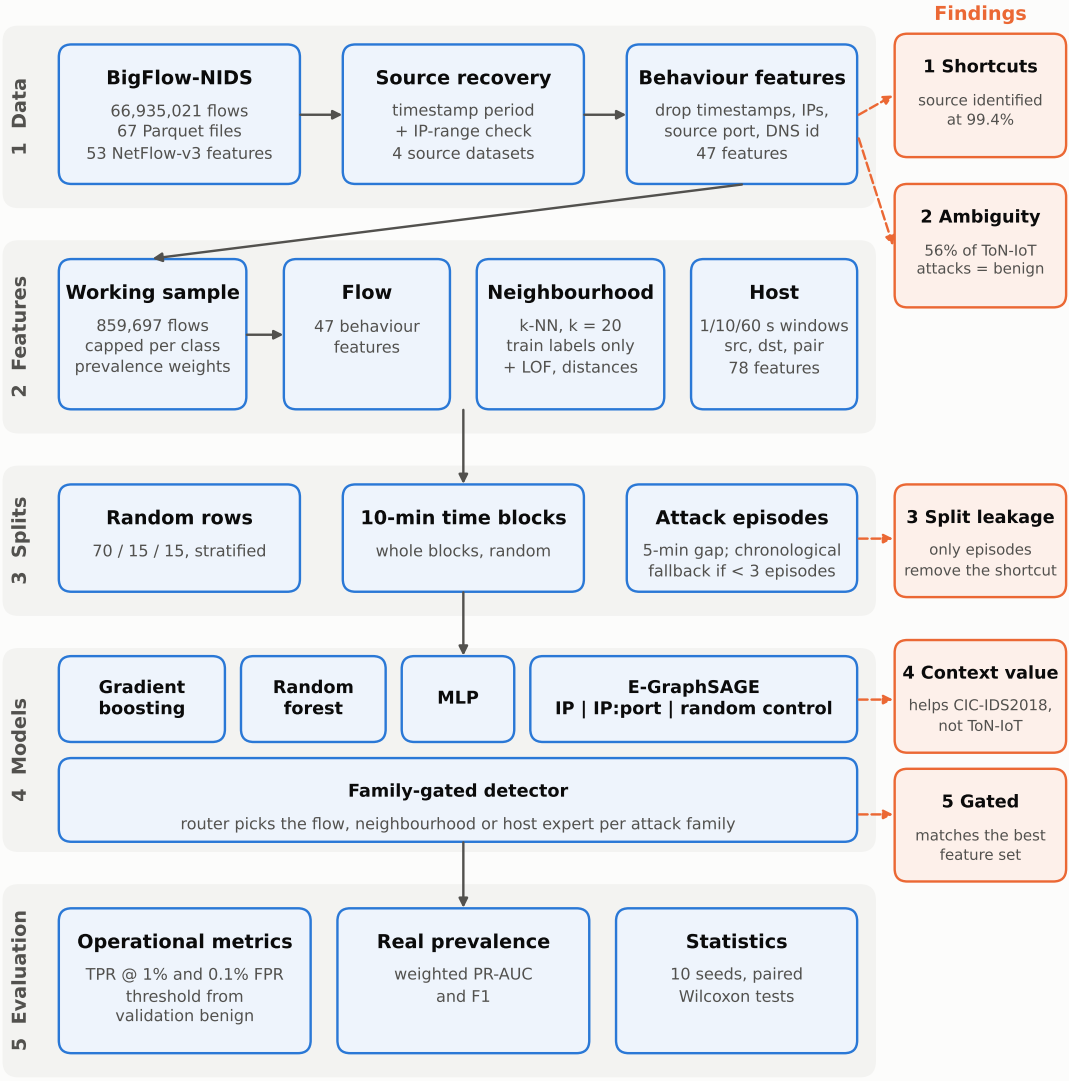
Findings
1 Data
BigFlow-NIDS
66,935,021 flows
67 Parquet files
53 NetFlow-v3 features
Source recovery
timestamp period
+ IP-range check
4 source datasets
Behaviour features
drop timestamps, IPs,
source port, DNS id
47 features
1 Shortcuts
source identified
at 99.4%
2 Ambiguity
56% of ToN-IoT
attacks = benign
2 Features
Working sample
859,697 flows
capped per class
prevalence weights
Flow
47 behaviour
features
Neighbourhood
k-NN, k = 20
train labels only
+ LOF, distances
Host
1/10/60 s windows
src, dst, pair
78 features
3 Splits
Random rows
70 / 15 / 15, stratified
10-min time blocks
whole blocks, random
Attack episodes
5-min gap; chronological
fallback if < 3 episodes
3 Split leakage
only episodes
remove the shortcut
4 Models
Gradient
boosting
Random
forest
MLP
E-GraphSAGE
IP | IP:port | random control
4 Context value
helps CIC-IDS2018,
not ToN-IoT
Family-gated detector
router picks the flow, neighbourhood or host expert per attack family
5 Gated
matches the best
feature set
5 Evaluation
Operational metrics
TPR @ 1% and 0.1% FPR
threshold from
validation benign
Real prevalence
weighted PR-AUC
and F1
Statistics
10 seeds, paired
Wilcoxon tests